# Database and DBMS Introduction

## What is a Database?

A Database is a reserved space in memory/storage where all the data of a particular application is stored in an organized manner so that it becomes easy to perform CRUD operations.

CRUD stands for:

- Create
- Read
- Update
- Delete

### Example

For a college management system, the database may store:

- Student Details
- Teacher Details
- Attendance Records
- Marks

All this data is stored in an organized way inside the database.

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## What is DBMS?

To perform CRUD operations directly on a database, we would need to write complex low-level code.

To simplify this process, we use a software called DBMS (Database Management System).

A DBMS is software that helps us create, manage, retrieve, update, and delete data from a database.

### Responsibilities of a DBMS

- Store Data
- Retrieve Data
- Update Data
- Delete Data
- Manage Security
- Handle Relationships

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## How Does DBMS Work?

A DBMS does not perform operations automatically.

To perform any operation, we must communicate with the DBMS using queries.

A Query is a request or instruction given to the DBMS.

These queries can be written in SQL programming language or JavaScript, depending on the type of DBMS being used.

For example:

### Simple SQL Query (MySQL)

```sql
SELECT * FROM students;
```

This query retrieves all records from the `students` table.

### Simple JavaScript Query Using Mongoose (MongoDB)

```javascript
const students = await Student.find();
```

This query retrieves all documents from the `students` collection.

### More Examples of Queries

Create Data

```sql
INSERT INTO students VALUES (1, 'Shahid');
```

Read Data

```sql
SELECT * FROM students;
```

Update Data

```sql
UPDATE students SET name='Ali' WHERE id=1;
```

Delete Data

```sql
DELETE FROM students WHERE id=1;
```

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# Types of DBMS

## 1. SQL DBMS (Relational Database Management System)

SQL DBMS stores data in tables consisting of rows and columns and uses SQL (Structured Query Language) to manage data.

#### Features

- Data is stored in Tables
- Uses SQL Queries
- Supports Relationships between tables
- Structured Schema

#### Examples

- MySQL
- PostgreSQL
- Oracle Database
- Microsoft SQL Server

#### Example Table

| ID | Name |
|---|---|
| 1 | Shahid |
| 2 | Ali |

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## 2. NoSQL DBMS

NoSQL DBMS stores data in a flexible format such as documents, key-value pairs, graphs, or columns.

#### Features

- Flexible Schema
- Stores Data in Documents, Collections, or Other Formats
- Suitable for Large-Scale Applications
- Easy to Scale

**Examples**

- MongoDB
- Cassandra
- Redis
- CouchDB

**Example Document (MongoDB)**

```
{
  "name": "Shahid",
  "age": 21
}
```

MongoDB stores data in collections and documents using a JSON-like format.

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## Summary

Database → Stores Data

DBMS → Manages the Database

Query → A request or instruction given to the DBMS to perform operations on data

Queries can be written in:

- SQL (for SQL DBMS like MySQL)
- JavaScript using Mongoose (for NoSQL DBMS like MongoDB)

### Types of DBMS

1. SQL DBMS

2. Examples: MySQL, PostgreSQL, Oracle, SQL Server

3. NoSQL DBMS

4. Examples: MongoDB, Cassandra, Redis, CouchDB